

ABE 721

Agricultural Systems Modelling and Simulation

Module 6: Advanced Applications in Agricultural Systems Modelling

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Chapter 6

Advanced Applications in Agricultural Systems Modelling

The preceding modules have established a rigorous theoretical foundation. Module 1 introduced CAD and systems logic. Module 2 translated realities into differential equations. Module 3 provided numerical techniques (Runge-Kutta, Finite Difference). Module 4 focused on model validation. Module 5 mapped the flows of mass and energy across spatial and temporal dimensions.

Module 6 represents the ultimate synthesis of these theories. We will explore advanced, highly sophisticated applications of agricultural modelling, drawing upon advanced texts in the course syllabus, including Peart and Curry (1998), Tayfur (2014) for soft computing, and Knudra (2000) for Computer-Aided Manufacturing (CAM).

6.1 6.1 Comprehensive Crop Simulation Modelling

Modern agricultural systems engineers utilize comprehensive, integrated crop simulation models like DSSAT and APSIM. These software suites consist of dozens of tightly coupled mechanistic sub-models representing phenology, carbon assimilation, and soil hydrology.

6.1.1 The Mathematics of Phenological Development

In mechanistic crop models, chronological time (days) is functionally useless because a maize plant grows significantly faster at $30°$C than it does at $15°$C. Models utilize **Thermal Time**, measured in **Growing Degree Days (GDD)**.

Thermal Time Formulation

Let T_{max} and T_{min} be daily temperatures. The basic daily accumulation (ΔTT) is:

$$\Delta TT = \max\left(0, \frac{T_{max} + T_{min}}{2} - T_{base}\right) \tag{6.1}$$

Advanced models utilize a truncated continuous integral over the diurnal temperature curve $T(t)$:

$$\Delta TT = \int_0^{24} f_{dev}(T(t))\, dt \tag{6.2}$$

Where the biological response function $f_{dev}(T)$ is:

$$f_{dev}(T) = \begin{cases} 0 & \text{if } T \leq T_{base} \\ \frac{T - T_{base}}{T_{opt} - T_{base}} & \text{if } T_{base} < T \leq T_{opt} \\ \frac{T_{crit} - T}{T_{crit} - T_{opt}} & \text{if } T_{opt} < T \leq T_{crit} \\ 0 & \text{if } T > T_{crit} \end{cases} \tag{6.3}$$

A specific growth stage is achieved when the integral of ΔTT reaches a genetic parameter (P_1).

$$\sum_{i=1}^{n} \Delta TT_i \geq P_1 \tag{6.4}$$

6.1.2 Biomass Assimilation and Partitioning

Let W be total dry biomass. The daily gross photosynthesis (P_{gross}) is:

$$P_{gross} = RUE \times PAR \times (1 - \exp(-k \cdot LAI)) \tag{6.5}$$

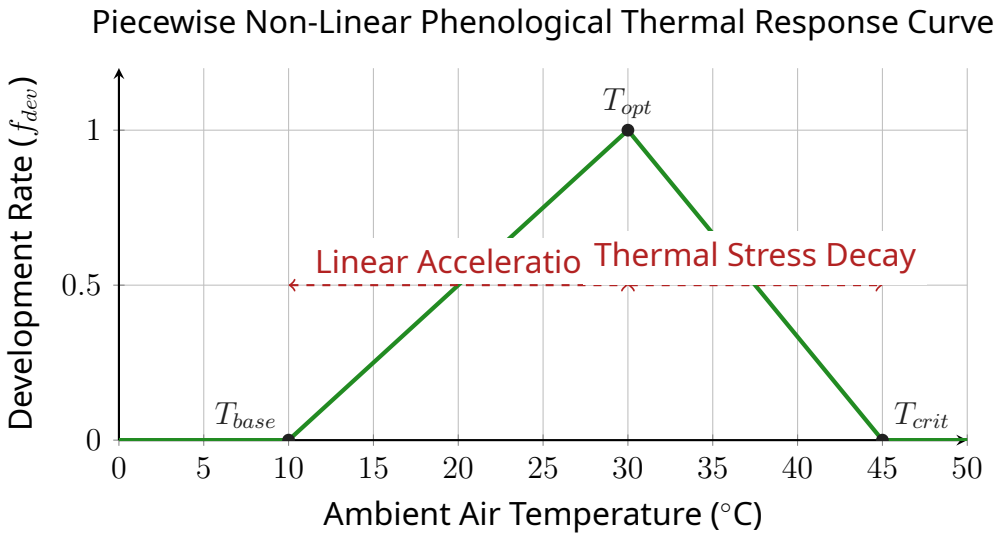


Figure 6.1: The biological development rate as a function of temperature.

The net daily accumulation of biomass is a strict mass-balance equation:

$$\frac{dW}{dt} = Y_g \left(P_{gross} - R_m \cdot W \right) \tag{6.6}$$

The Partitioning Matrix

The total newly generated biomass (ΔW) must be dynamically distributed to different anatomical organs using a partitioning vector $\alpha(t)$:

$$\alpha(t) = \begin{bmatrix} \alpha_{root}(t) \\ \alpha_{stem}(t) \\ \alpha_{leaf}(t) \\ \alpha_{grain}(t) \end{bmatrix}, \quad \text{subject to} \quad \sum \alpha_i(t) = 1 \tag{6.7}$$

6.2 6.2 Spatial Modelling and GIS Integration

6.2.1 Mathematical Representation of Spatial Data

In computational systems, continuous geographic reality must be discretized.

1. Vector Data Architecture

Vector systems represent space using discrete geometric primitives defined by coordinate matrices:

$$\mathbf{P} = \begin{bmatrix} x_1 & y_1 \\ x_2 & y_2 \\ \vdots & \vdots \\ x_n & y_n \\ x_1 & y_1 \end{bmatrix} \tag{6.8}$$

Area is calculated using the Shoelace Formula:

$$Area = \frac{1}{2} \left| \sum_{i=1}^{n-1} (x_i y_{i+1} - x_{i+1} y_i) \right| \tag{6.9}$$

2. Raster Data Architecture

Raster systems discretize space into a regular 2D matrix of cells (pixels). Let **R** be an $M \times N$ matrix. The tremendous power of raster GIS lies in Map Algebra. Calculating a crop stress index map (**S**) by subtracting a water demand raster (**D**) from a soil moisture supply raster (**M**):

$$\mathbf{S} = \mathbf{M} - \mathbf{D} \implies S_{i,j} = M_{i,j} - D_{i,j} \quad \forall i, j \tag{6.10}$$

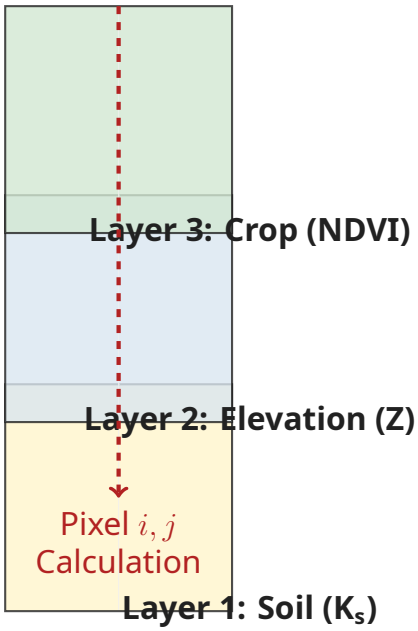


Figure 6.2: Distributed spatial modelling utilizing GIS Map Algebra.

6.2.2 Distributed Spatial Modelling: The USLE Case Study

The Universal Soil Loss Equation (USLE) is calculated using raster map algebra for every single pixel in the watershed matrix:

$$A(x,y) = R(x,y) \cdot K(x,y) \cdot LS(x,y) \cdot C(x,y) \cdot P(x,y) \tag{6.11}$$

6.2.3 Spatial Interpolation: The Mathematics of Kriging

The estimated value $\hat{Z}$ at an unsampled location $\mathbf{s}_0$ is calculated as a linearly weighted sum of the n known observations:

$$\hat{Z}(\mathbf{s}_0) = \sum_{i=1}^{n} \lambda_i Z(\mathbf{s}_i) \tag{6.12}$$

The optimal weights λ_i are found by solving a spatial covariance matrix:

$$\begin{bmatrix} \gamma_{11} & \cdots & \gamma_{1n} & 1 \\ \vdots & \ddots & \vdots & \vdots \\ \gamma_{n1} & \cdots & \gamma_{nn} & 1 \\ 1 & \cdots & 1 & 0 \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \vdots \\ \lambda_n \\ \mu \end{bmatrix} = \begin{bmatrix} \gamma_{10} \\ \vdots \\ \gamma_{n0} \\ 1 \end{bmatrix} \tag{6.13}$$

6.3 6.3 Soft Computing: Artificial Neural Networks in Agriculture

As outlined by Tayfur (2014), *Soft Computing in Water Resources Engineering*, highly complex agricultural systems frequently defy strict mechanistic PDE modelling. Engineers deploy **Artificial Neural Networks (ANNs)**.

6.3.1 Architecture of the Multi-Layer Perceptron (MLP)

The raw linear summation (z_j) arriving at a neuron is the dot product of the inputs and weights:

$$z_j = \sum_{i=1}^{n} w_{ij} x_i + b_j = \mathbf{w}_j^T \mathbf{x} + b_j \tag{6.14}$$

This is passed through a non-linear **Activation Function** $f(z)$ (e.g., Sigmoid):

$$a_j = \sigma(z_j) = \frac{1}{1 + e^{-z_j}} \tag{6.15}$$

6.3.2 Training the Network: The Backpropagation Algorithm

The objective is to minimize the global Cost Function $J(\mathbf{W}, \mathbf{b})$:

$$J(\mathbf{W}, \mathbf{b}) = \frac{1}{2m} \sum_{k=1}^{m} \left(\hat{y}^{(k)} - y_{true}^{(k)} \right)^2 \tag{6.16}$$

Using the Chain Rule, the gradient for a specific weight is propagated backwards:

$$\frac{\partial J}{\partial w_{ij}^{(l)}} = \frac{\partial J}{\partial a_j^{(l)}} \times \frac{\partial a_j^{(l)}}{\partial z_j^{(l)}} \times \frac{\partial z_j^{(l)}}{\partial w_{ij}^{(l)}} = \delta_j^{(l)} \cdot a_i^{(l-1)} \tag{6.17}$$

The network updates its weights using Gradient Descent:

$$w_{ij}^{(new)} = w_{ij}^{(old)} - \alpha \frac{\partial J}{\partial w_{ij}} \tag{6.18}$$

6.4 6.4 Computer-Aided Manufacturing (CAM) and CNC Integration

Referencing Knudra (2000), the digital model must be translated into physical steel using CAM and Computer Numerical Control (CNC) machining via **G-Code**.

6.4.1 Kinematics and Transformation Matrices

Engineers handle complex 3D kinematics using **Homogeneous Transformation Matrices** (4×4 matrices) combining translation and rotation:

$$\mathbf{T} = \begin{bmatrix} R_{11} & R_{12} & R_{13} & T_x \\ R_{21} & R_{22} & R_{23} & T_y \\ R_{31} & R_{32} & R_{33} & T_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{6.19}$$

For example, a pure rotational movement around the Z-axis by an angle θ:

$$\mathbf{R}_z(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 & 0 \\ \sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{6.20}$$

6.5 6.5 Bio-Economic Optimization and Decision Support Systems

A commercial farm manager must make critical decisions regarding resource allocation to absolutely maximize global financial profit using a **Linear Programming (LP)** model.

Standard Mathematical Formulation

The **Objective Function** to strictly maximize total farm profit (Z) is:

$$\text{Maximize } Z = \sum_{j=1}^{n} c_j x_j = \mathbf{c}^T \mathbf{x} \tag{6.21}$$

Subject to constraints:

$$\sum_{j=1}^{n} a_{ij} x_j \leq b_i \quad \text{for } i = 1, 2, \ldots, m \tag{6.22}$$

$$x_j \geq 0 \quad \text{for all } j \tag{6.23}$$

In standard matrix notation:

$$\begin{aligned} \text{Maximize} \quad & Z = \mathbf{c}^T \mathbf{x} && (6.24) \\ \text{Subject to} \quad & \mathbf{A}\mathbf{x} \leq \mathbf{b} && (6.25) \\ & \mathbf{x} \geq \mathbf{0} && (6.26) \end{aligned}$$

Shadow Prices and Dual Optimization

Solving the LP simultaneously solves the **Dual Problem**, yielding the *Shadow Price* (λ_i) for every single constrained resource:

$$\lambda_i = \frac{\partial Z^*}{\partial b_i} \tag{6.27}$$

End of Module 6 and the ABE 721 Course Notes Suite.